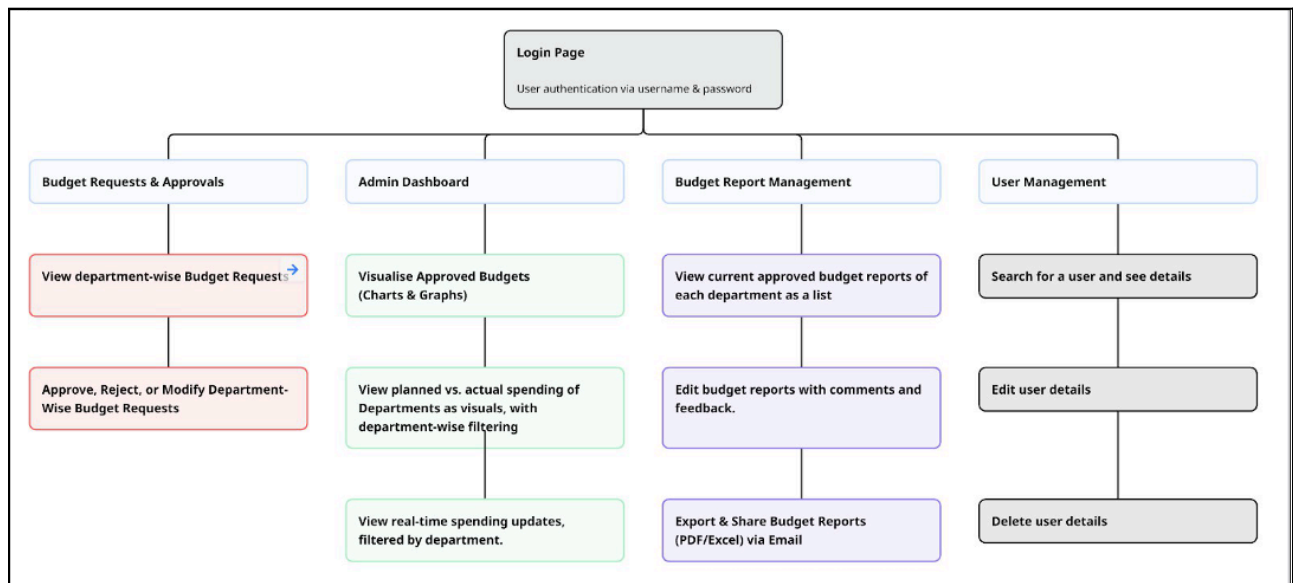


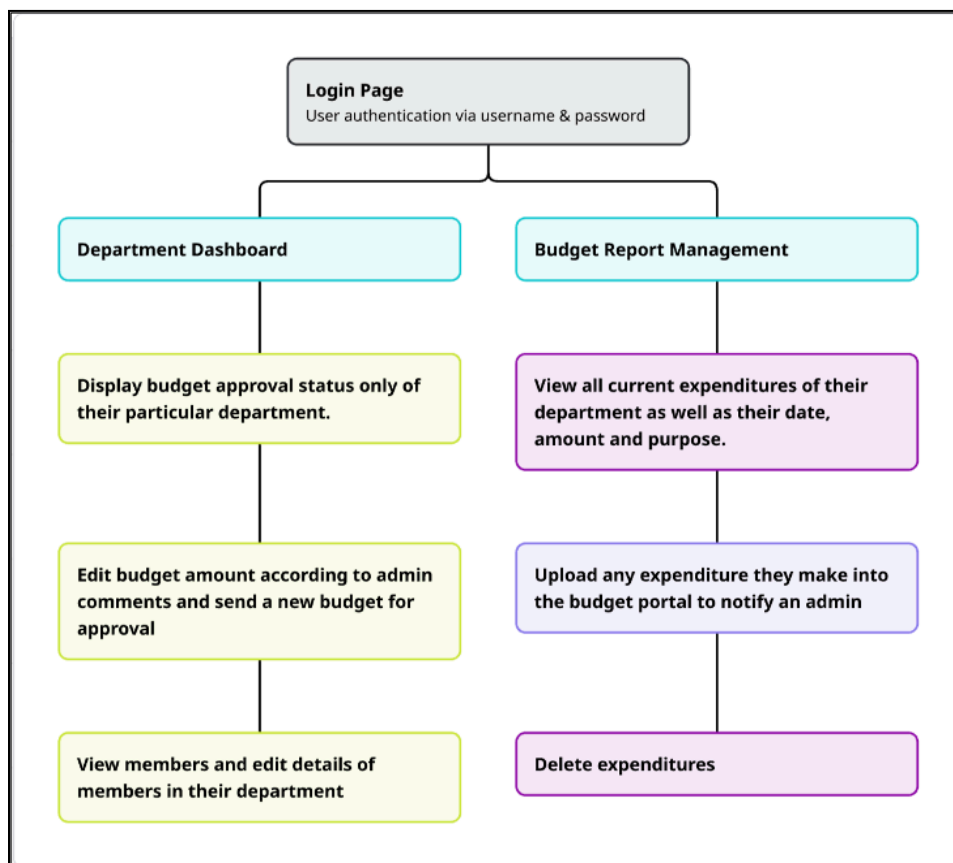
Computer Science Internal Assessment Criterion B - Solution Overview

Pages Overview

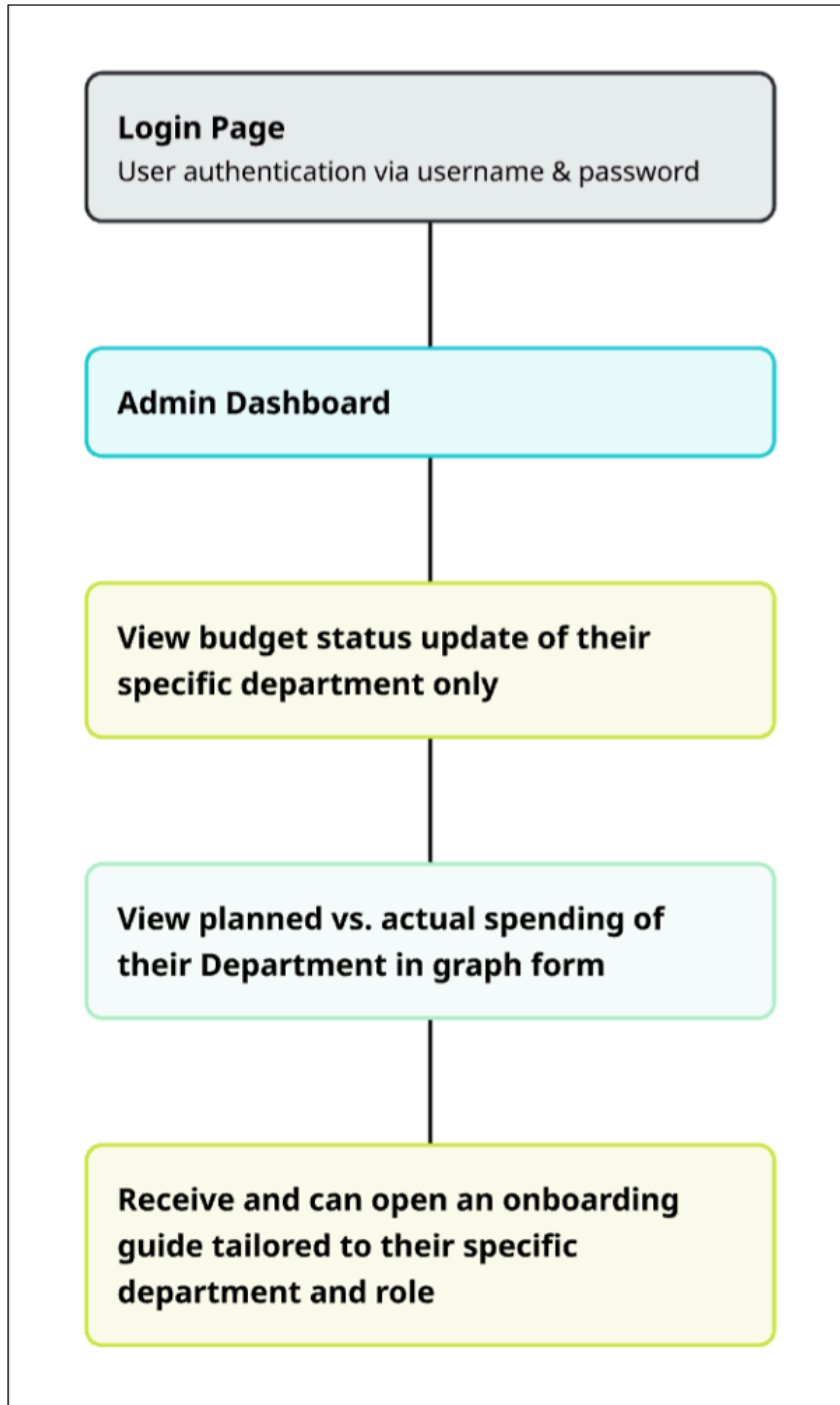
1. Admin pages :



2. Department Head pages



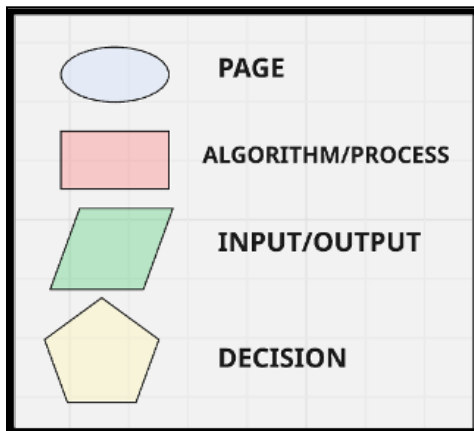
3. General Employee pages :



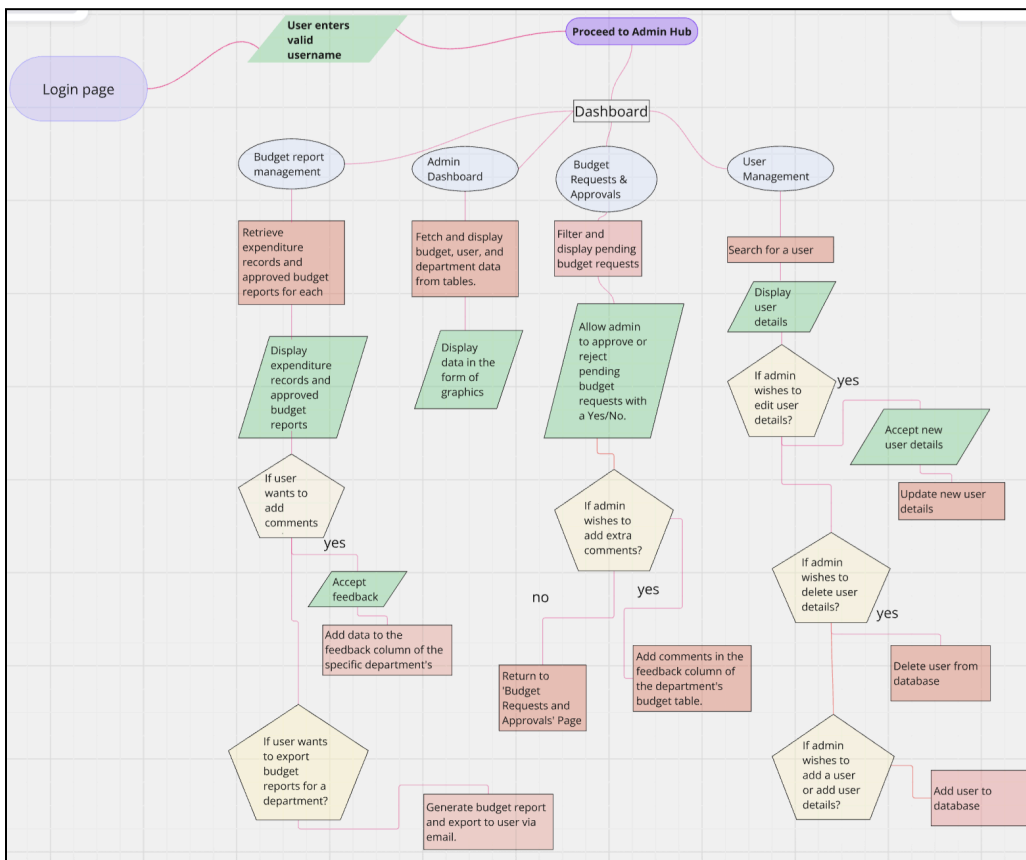
System flowchart

- Subjected to vary from final product
- Note: In some flowcharts, decision nodes include only a “Yes” pathway because the condition represents a user-initiated action (e.g., clicking a button). If the action is not performed, no process is triggered, so a “No” pathway is not required.

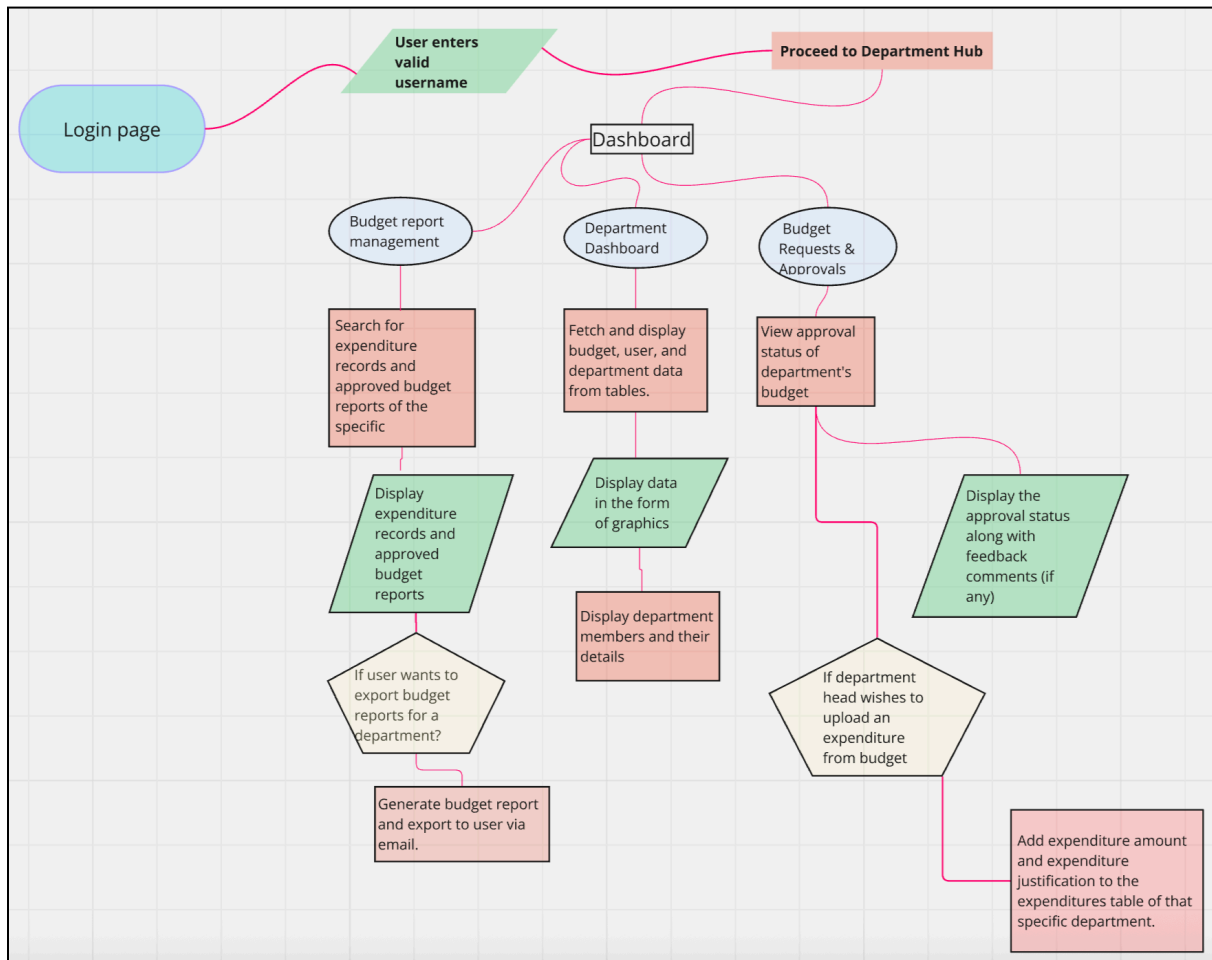
KEY -



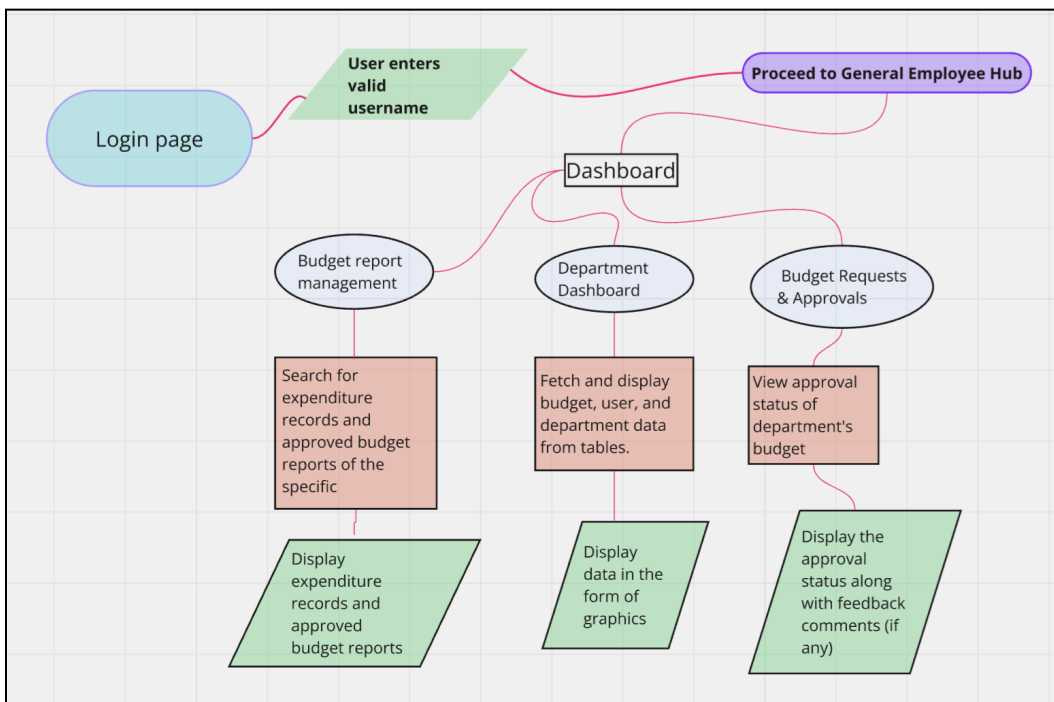
1. Admin side :



2. Department Head Side :



3. General Employees Side :



User Interface + Flowchart of all features

1. Login Page (visible to all parties)

admin, employees and department heads see this page.

only authorised users can access the web-site.

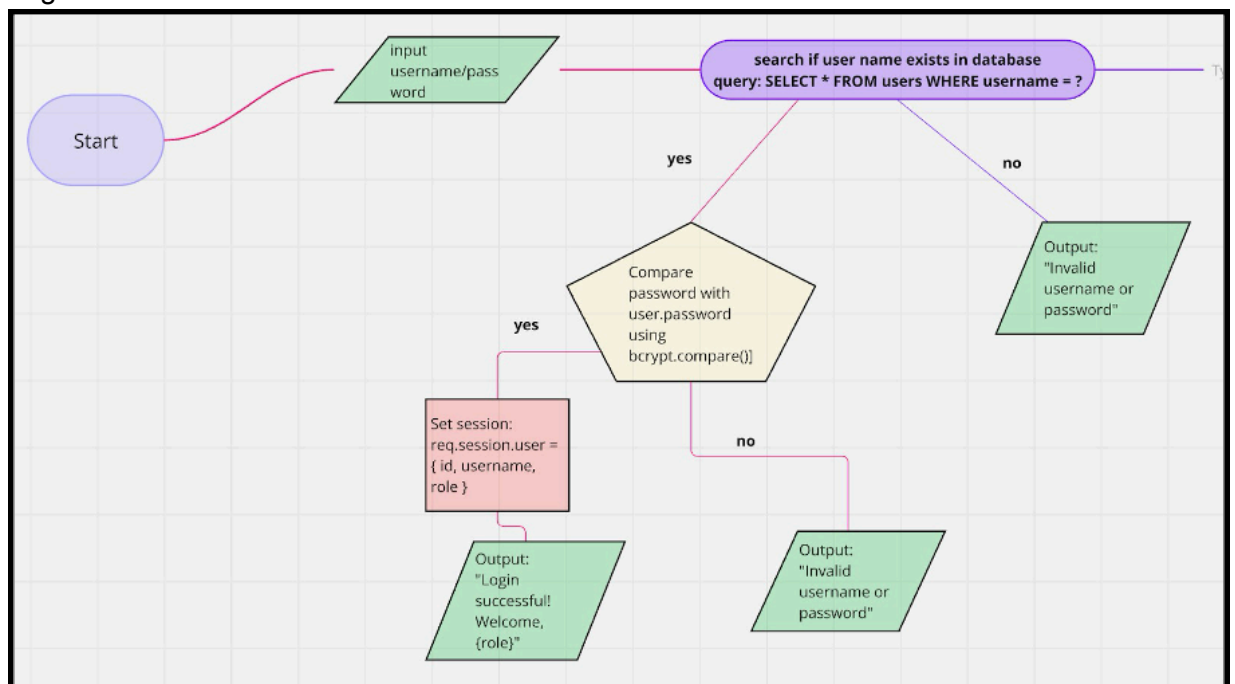
LOGIN PAGE

USERNAME

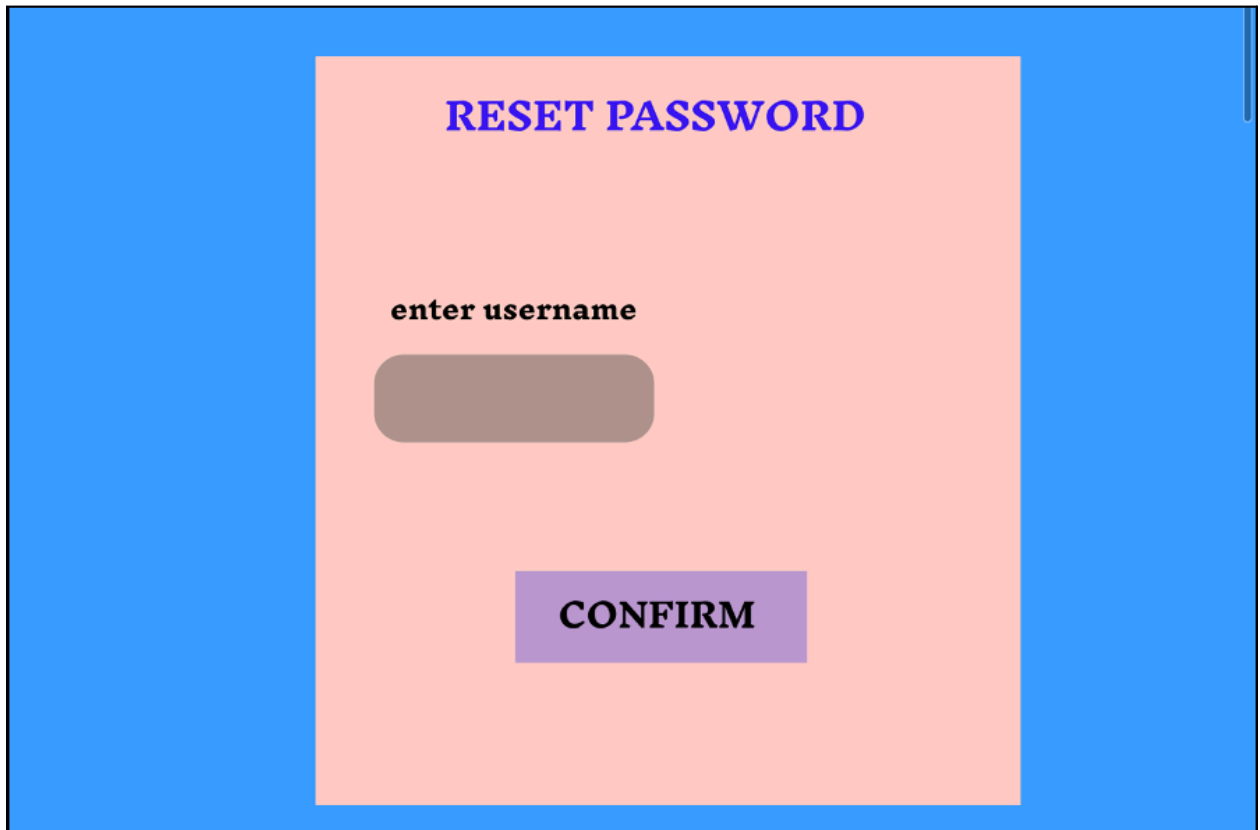
PASSWORD

[reset password?](#)

Login Process :



2. Reset Password Page (visible to all parties)

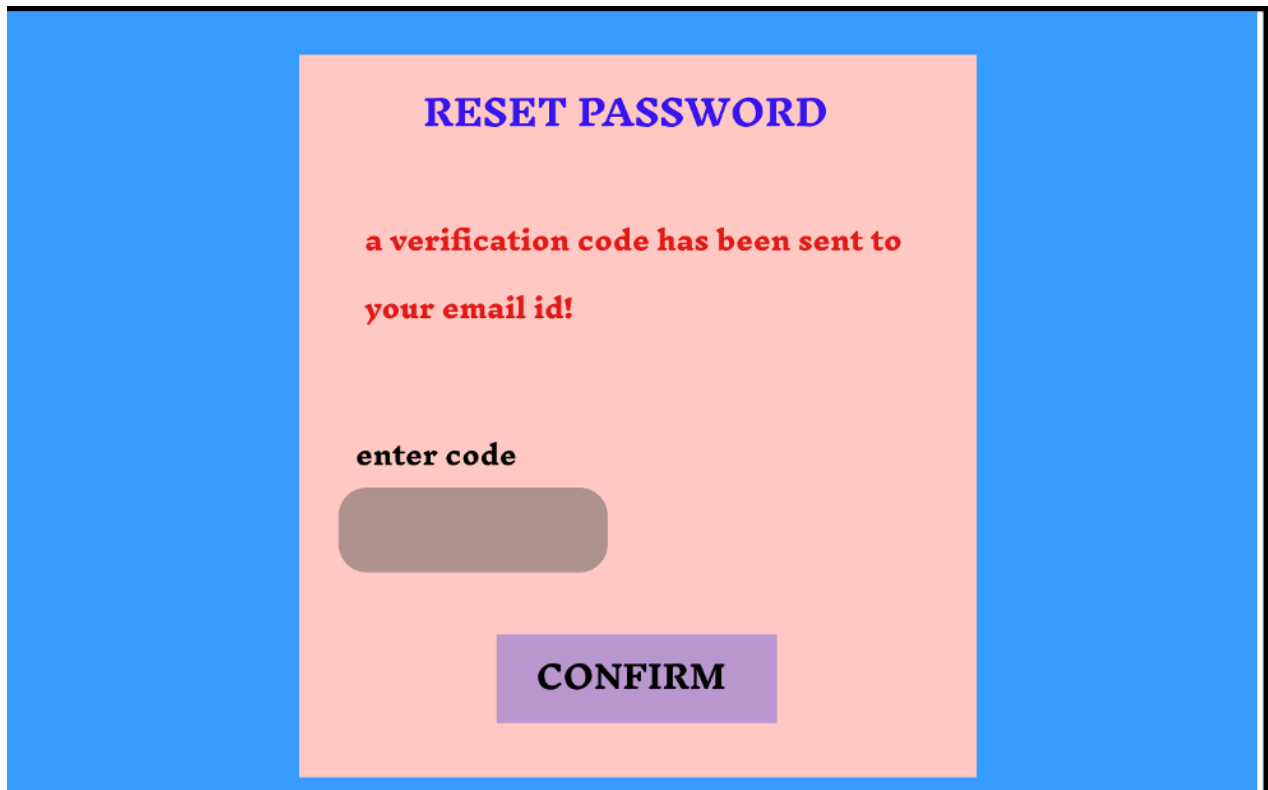


The image shows a web page with a blue background. In the center is a light pink rectangular box. At the top of this box, the text "RESET PASSWORD" is written in a bold, blue, serif font. Below this, the text "enter username" is written in a bold, black, sans-serif font. Underneath the text is a light brown, rounded rectangular input field. At the bottom of the pink box is a purple rectangular button with the word "CONFIRM" written in a bold, black, sans-serif font.

RESET PASSWORD

enter username

CONFIRM



The image shows a web page with a blue background. In the center is a light pink rectangular box. At the top of this box, the text "RESET PASSWORD" is written in a bold, blue, serif font. Below this, the text "a verification code has been sent to your email id!" is written in a bold, red, sans-serif font. Underneath the text is the text "enter code" in a bold, black, sans-serif font. Below "enter code" is a light brown, rounded rectangular input field. At the bottom of the pink box is a purple rectangular button with the word "CONFIRM" written in a bold, black, sans-serif font.

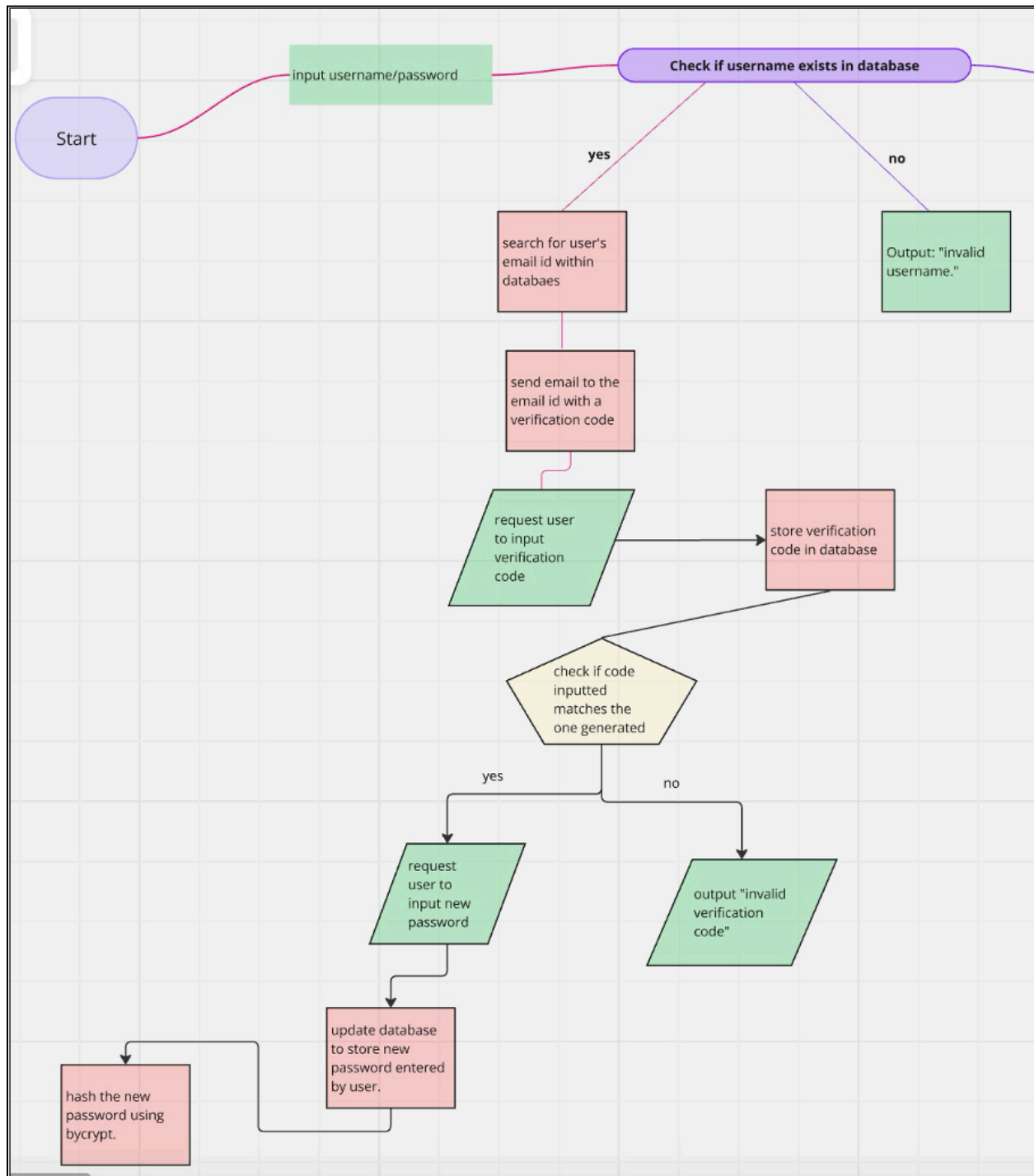
RESET PASSWORD

**a verification code has been sent to
your email id!**

enter code

CONFIRM

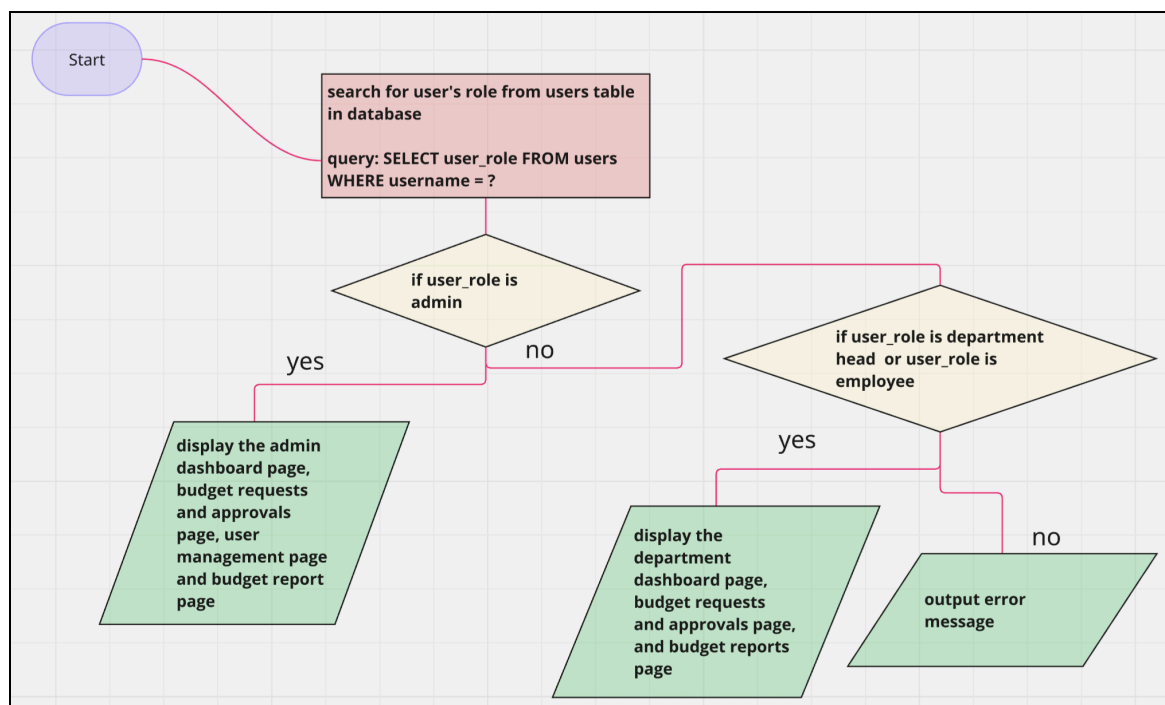
Password reset process :



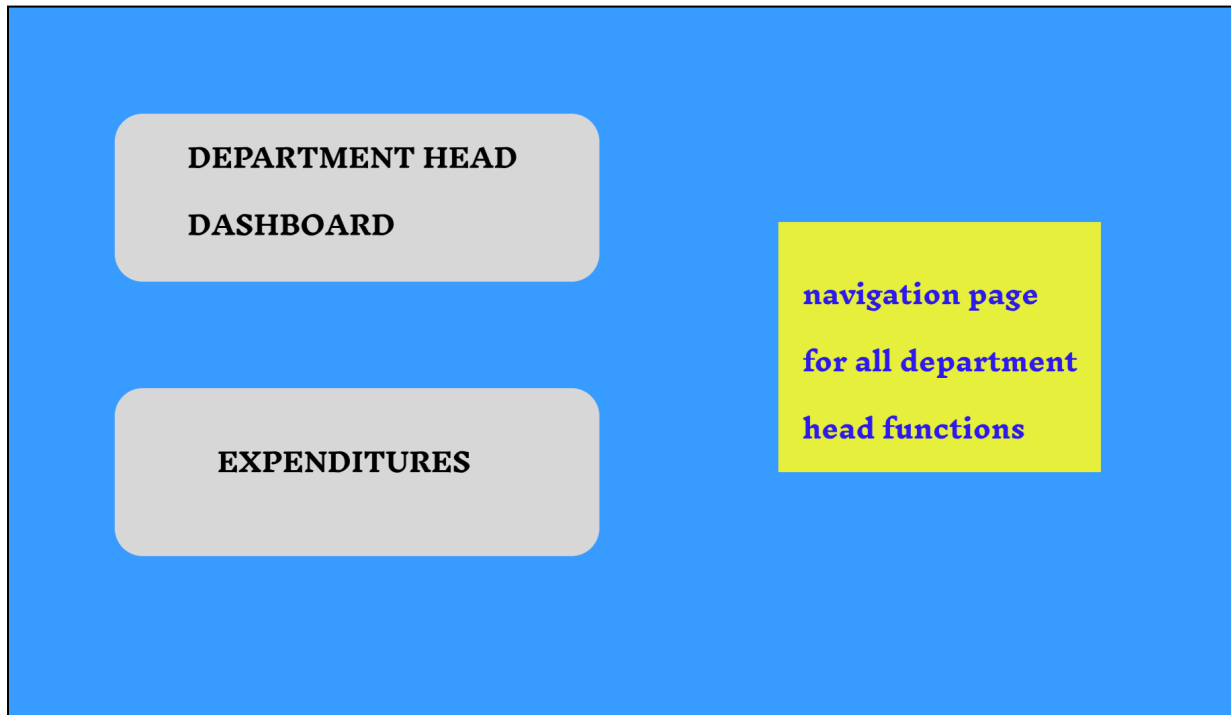
3. Admin Functions Page (visible to admins when they log in)



Admin Page verification functionality :



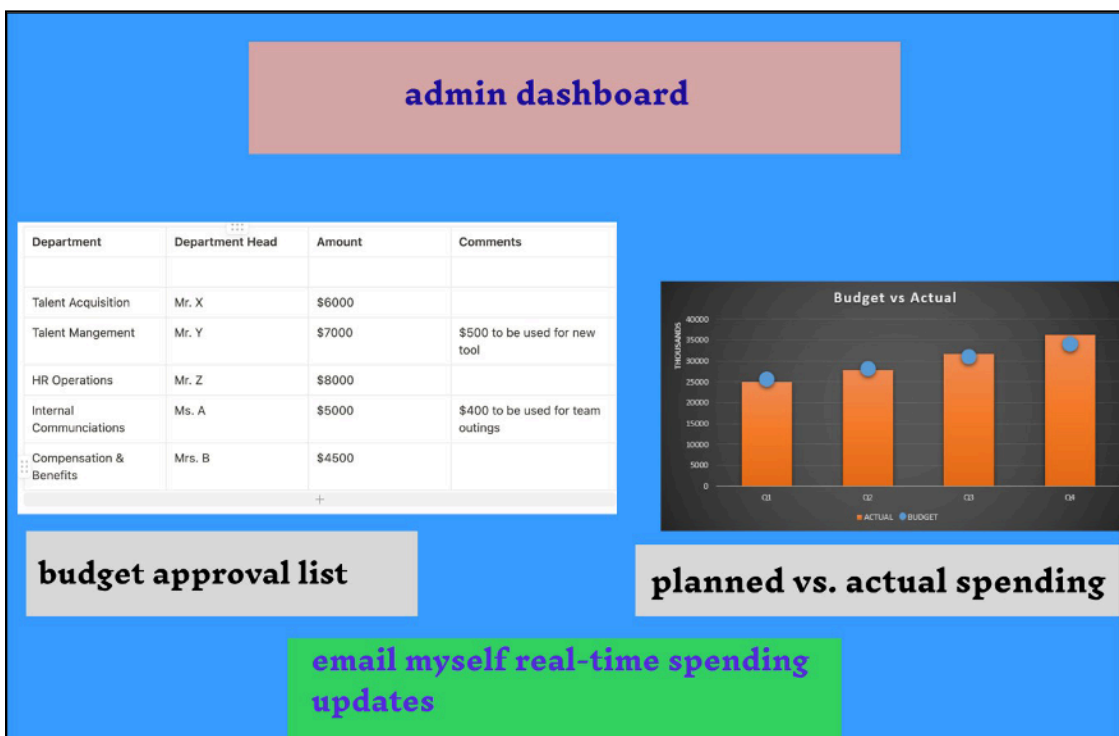
4. Department Head Functions Page (visible to department heads)



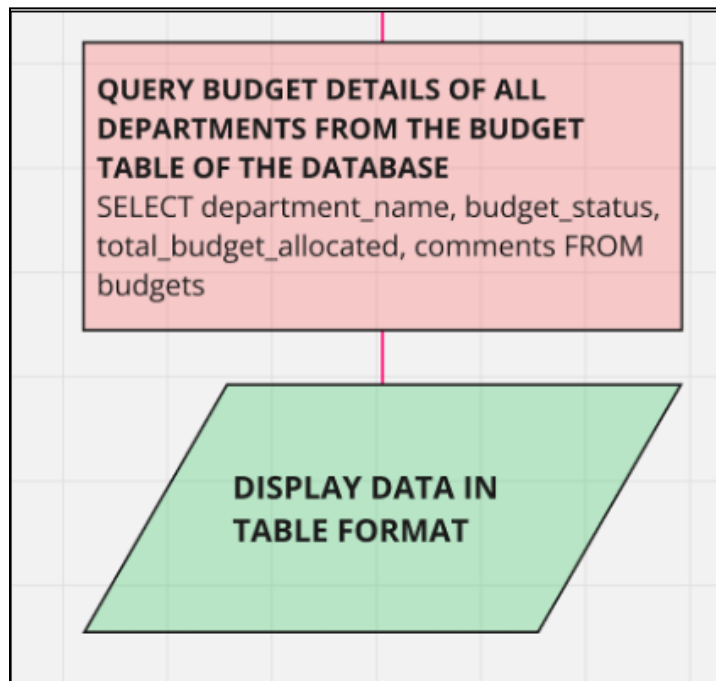
General Employees do not have a functions page as they only have 1 main dashboard page; their interface will hence be displayed later.

Admins :

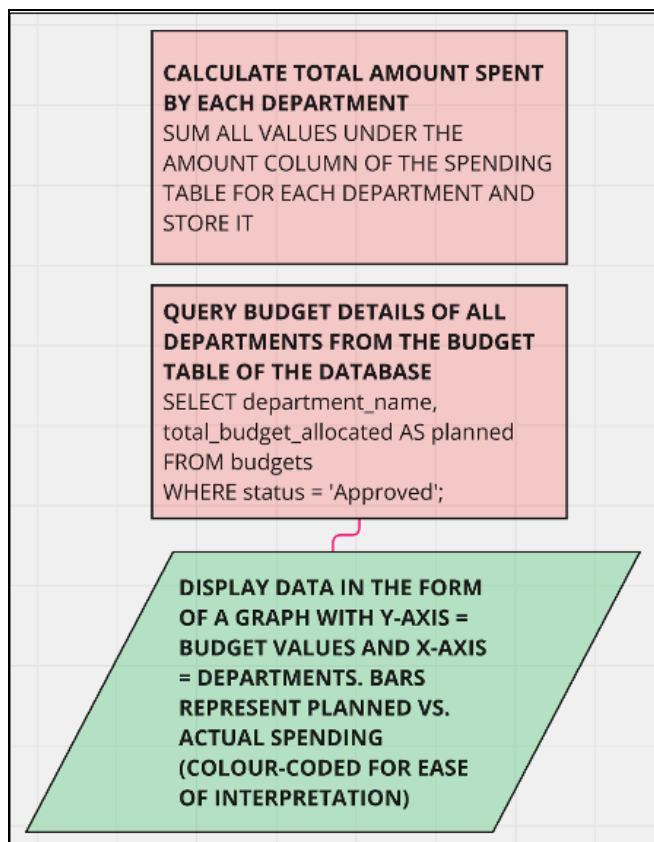
1. Admin Dashboard Page



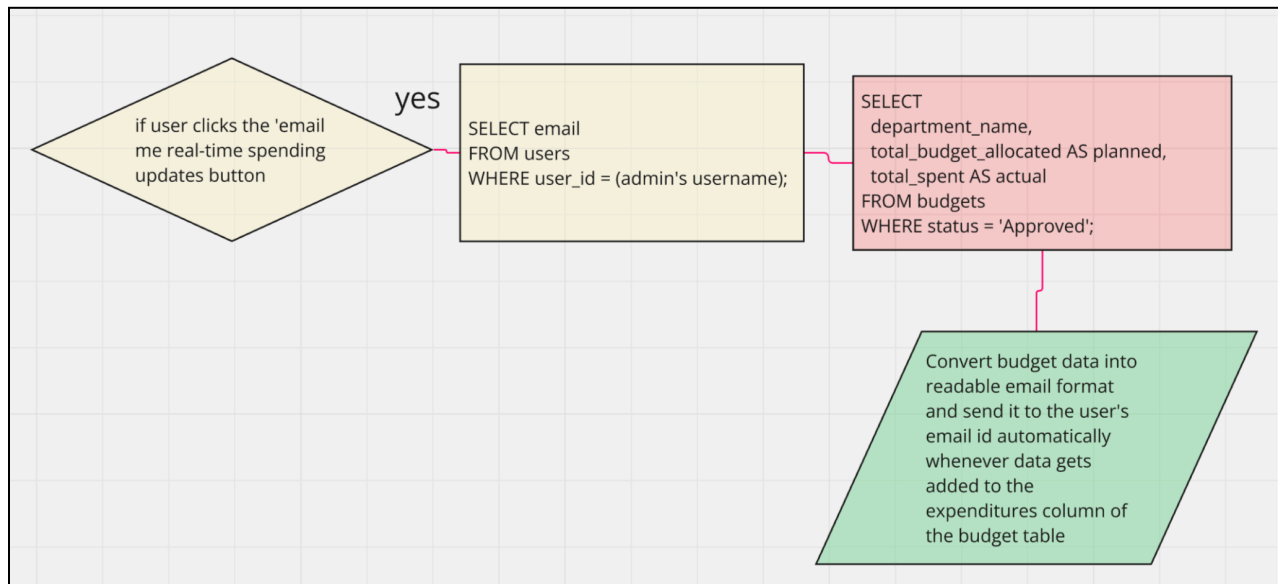
Process for displaying the budget approval list : ‘



Process for displaying the planned-vs-actual spending graph -



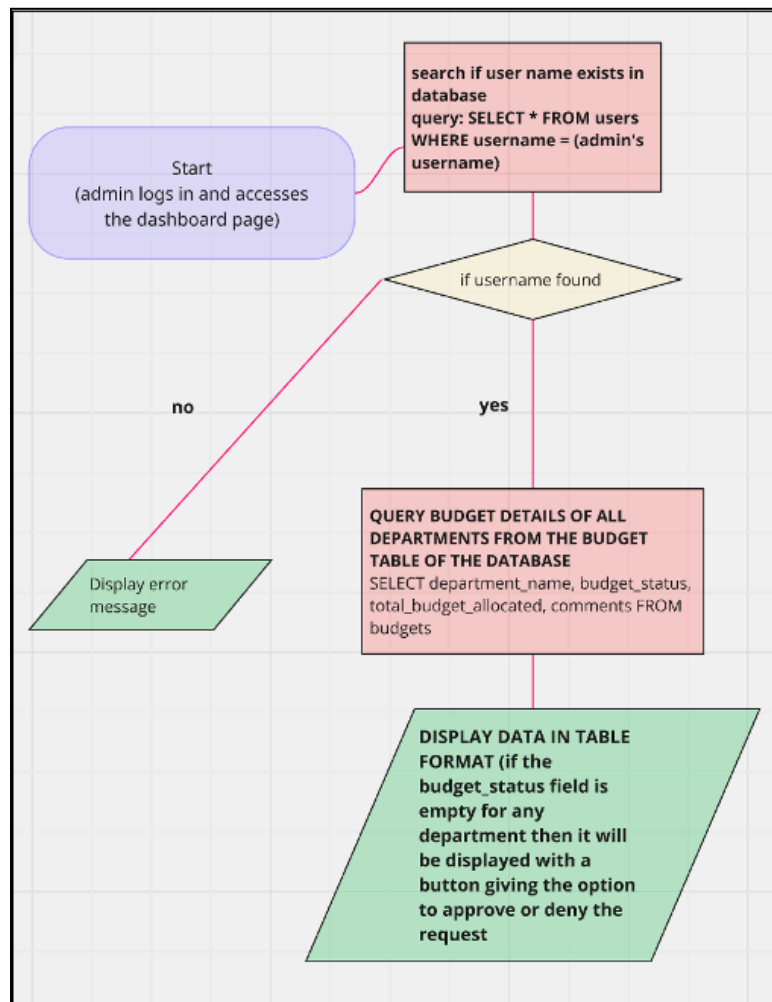
Process for emailing real-time spending updates :



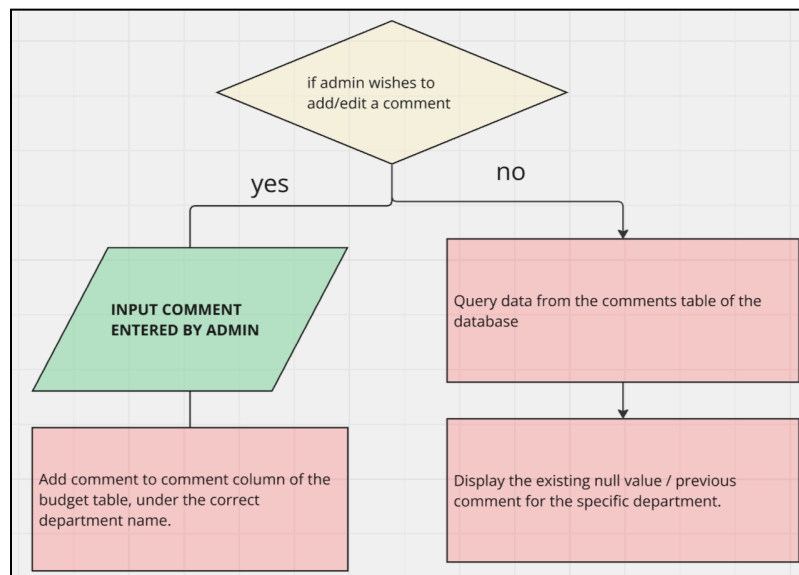
Budget Requests and Approvals Page:

budget requests and approvals				
Department	Department Head	Amount	Approve/ Deny	Comments
Talent Acquisition	Mr. X	\$6000	A	Add comment?
Talent Mangement	Mr. Y	\$7000	D	\$500 to be used for new tool
HR Operations	Mr. Z	\$8000	D	Add comment?
Internal Communciations	Ms. A	\$5000	A	\$400 to be used for team outings
Compensation & Benefits	Mrs. B	\$4500	A	Add comment?
+				

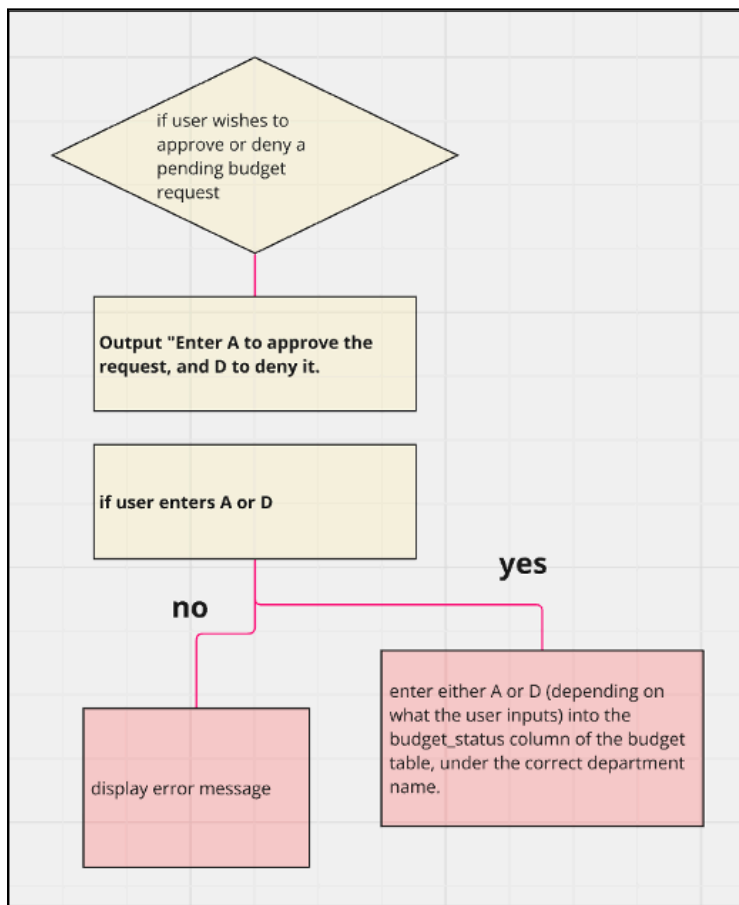
Process for displaying the departmental budget request list -



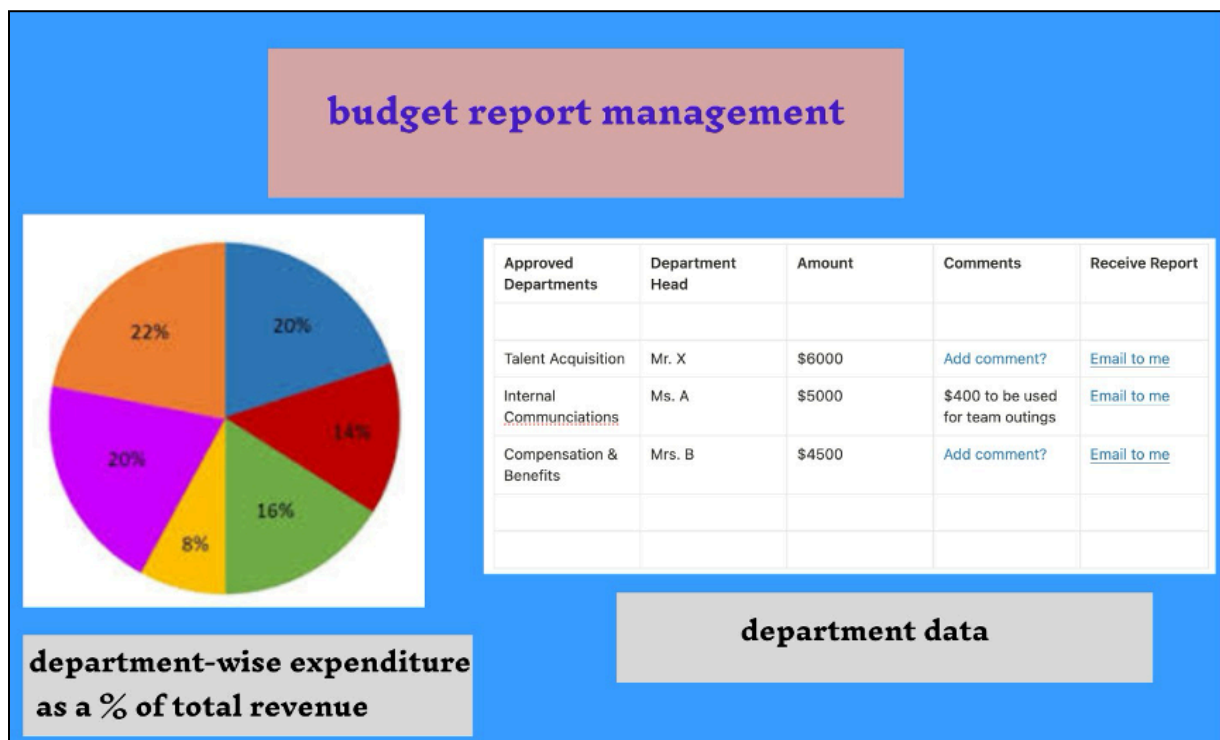
Process of adding a comment -



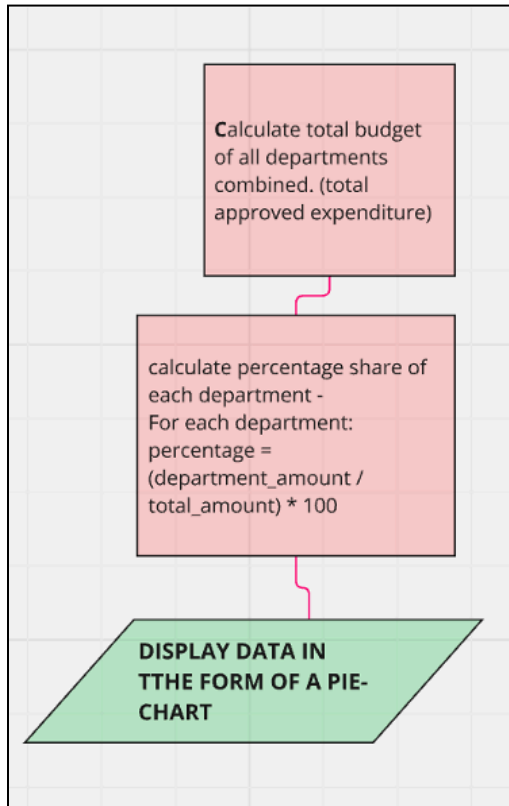
Process of admin approving/denying a budget request -



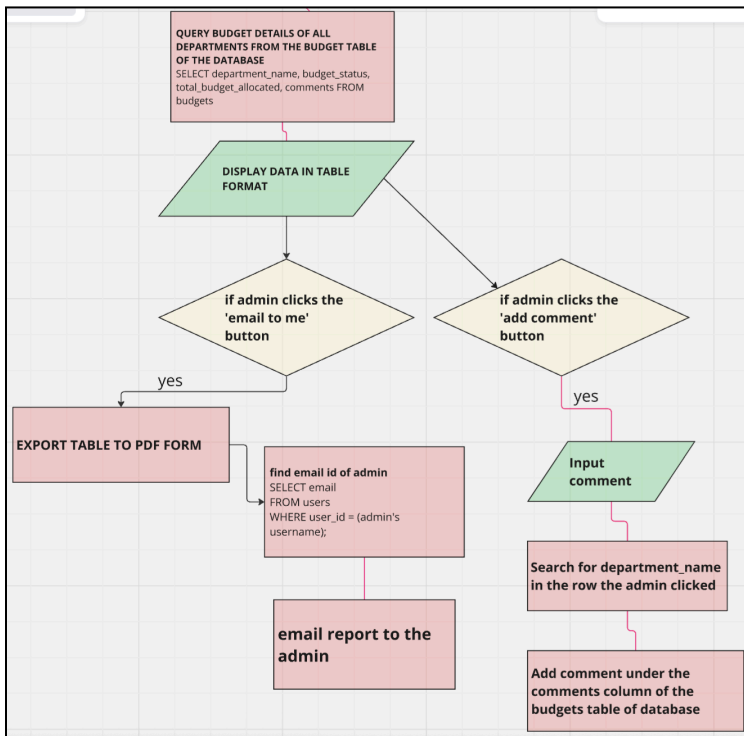
Budget Report Management Page



Process of displaying the pie-chart -



Process for the 'email to me' and 'add comments' features -



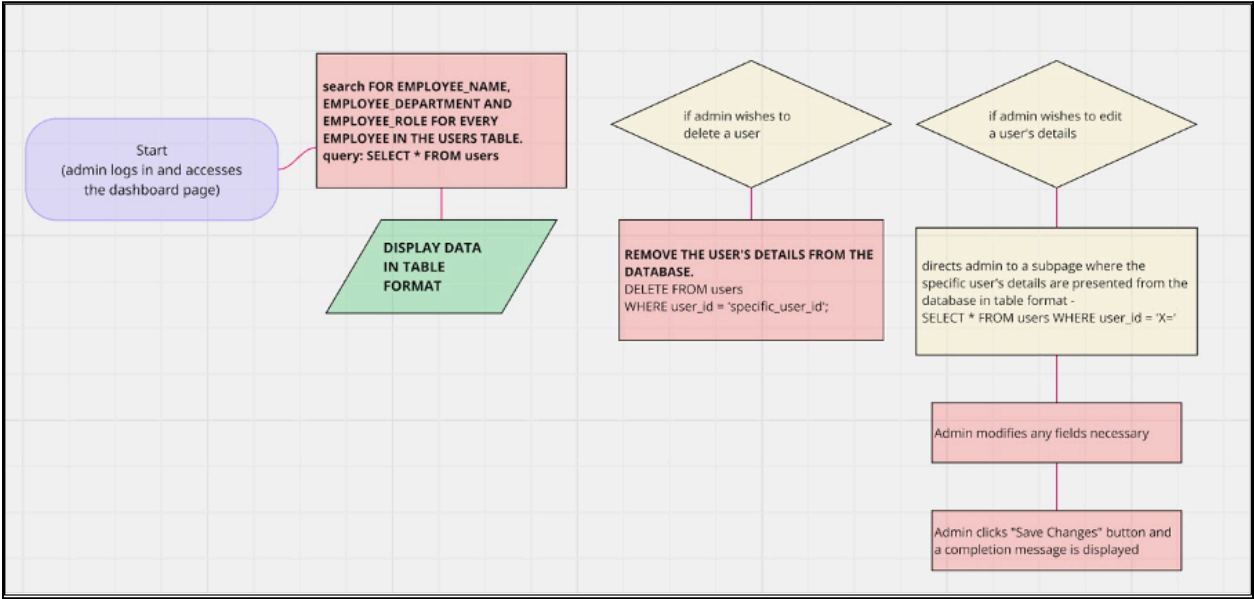
User Management Page

user management

search?

Employee Name	Employee Department	Employee Role	Edit	Remove
Mrs. Y	Talent <u>Acquisiton</u>	Secretary	Edit?	Delete
Mr. X	Talent Acquisition	Department Head	Edit?	Delete
Mrs. B	Compensation & Benefits	Department Head	Edit?	Delete
Mr. N	Internal Communications	Assistant	Edit?	Delete

User management functionality -



Department heads :

Department Head Dashboard Page -

department dashboard

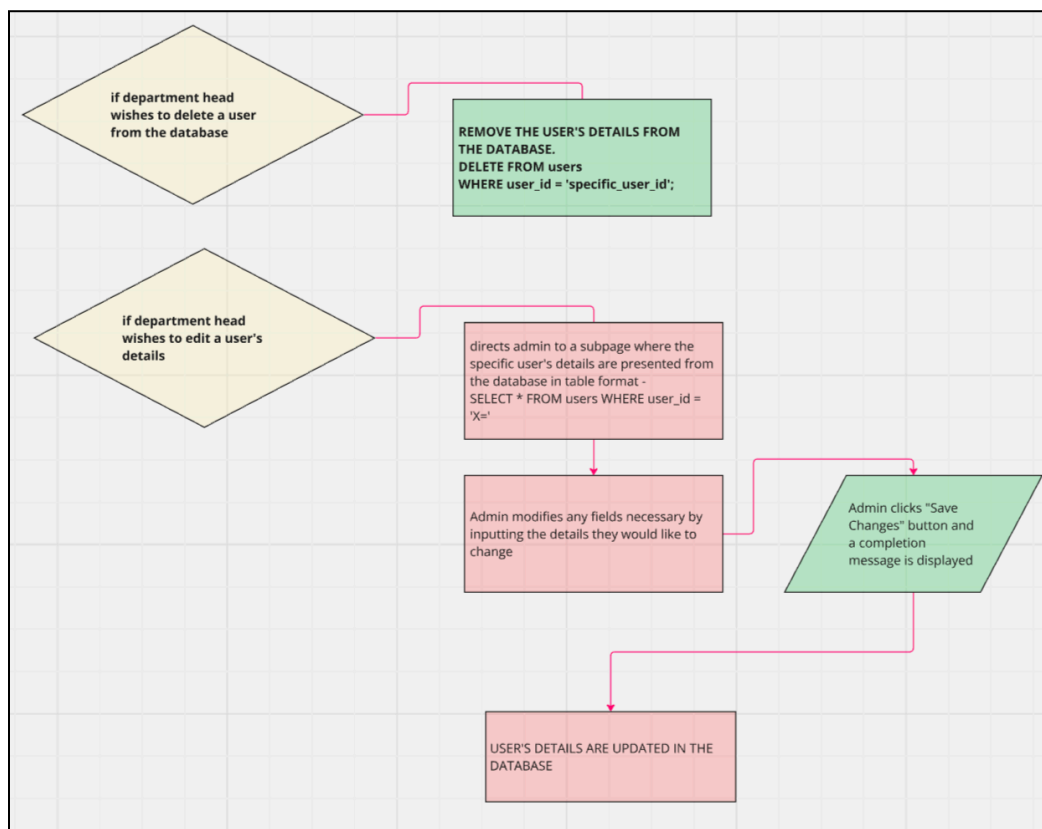
BUDGET APPROVAL STATUS -

Department	Department Head	Amount	Approve/ Deny	Comments
Talent Management	Mr. Y	\$7000	D	\$500 to be used for new tool

employee details ->

Employee Name	Employee Department	Employee Role	Edit	Remove
Mrs. Y	Talent Acquisition	Secretary	Edit?	Delete
Mr. X	Talent Acquisition	Department Head	Edit?	Delete
Mrs. B	Talent Acquisition	Analyst	Edit?	Delete
Mr. N	Talent Acquisition	Assistant	Edit?	Delete

User functionalities-



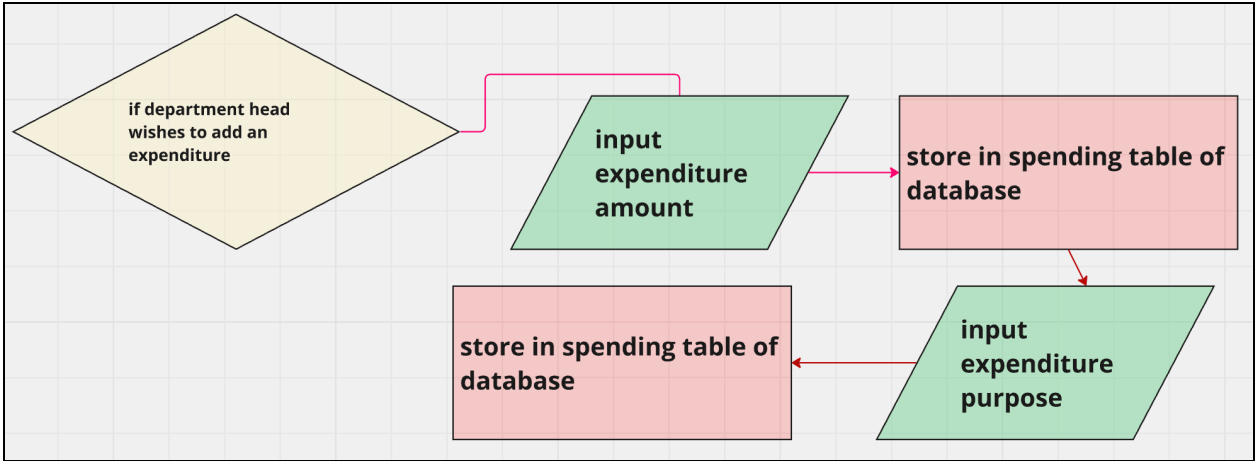
Department Head Expenditure Tracker

department expenses

Serial no.	Expenditure amount	Expenditure purpose	Remove
1	\$5000	Company-sponsored trip	
2	\$300	new laptops for HR team	Delete
3	\$890	AI tool for task tracking	Delete
	Add expenditure amount	Add expenditure purpose	Delete

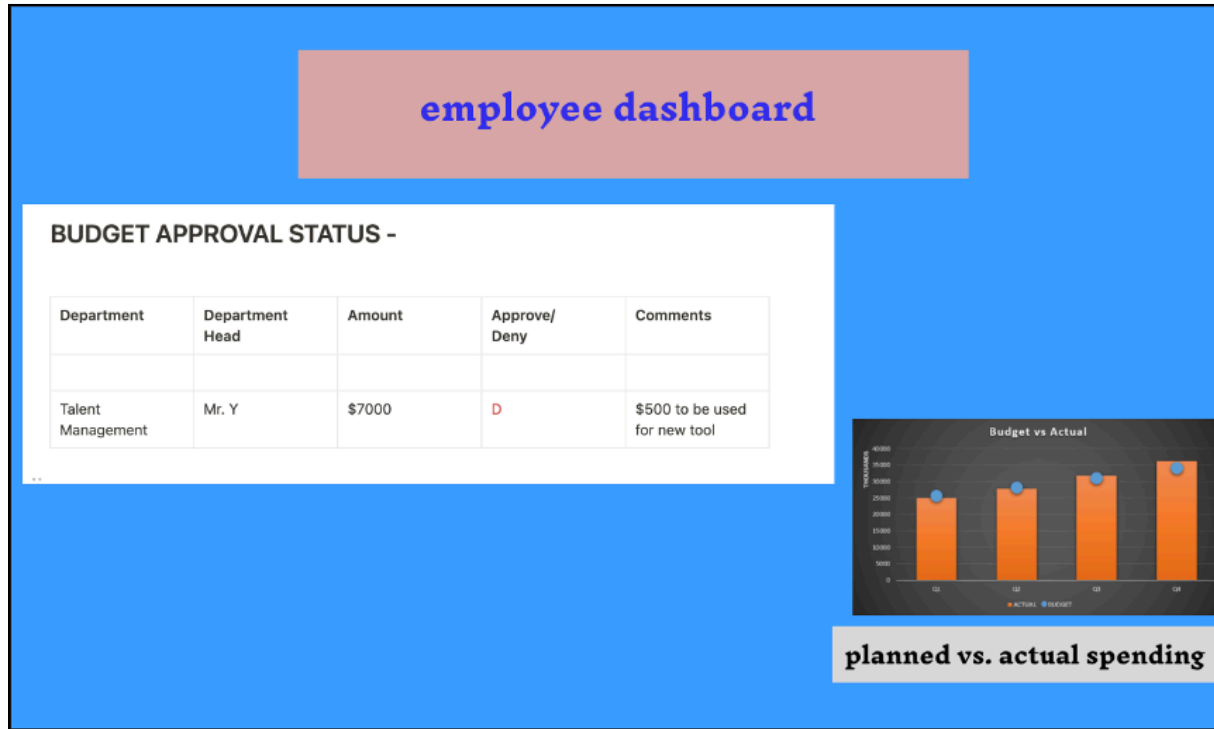
expenditure tracker

Expenditure functionality -



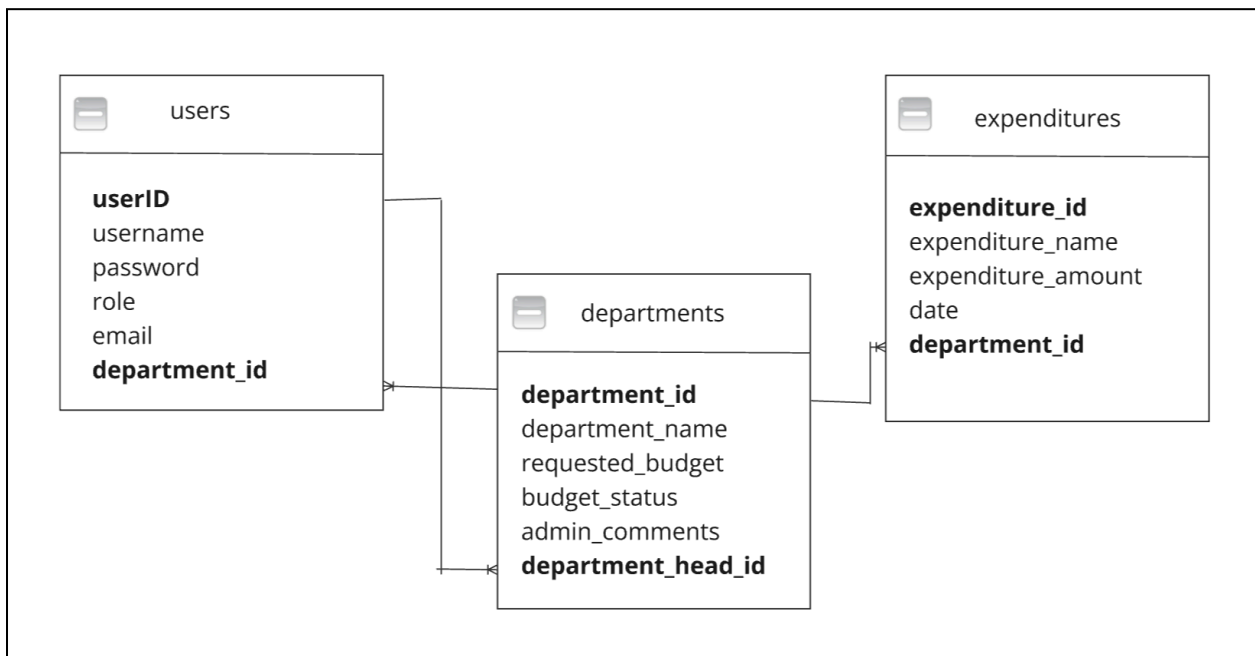
General Employees :

Employee Dashboard Page



Database Design

Entity relationship overview/diagram:



Users Table :

The database stores information about the admins, department heads and general employees in the company to successfully authenticate users, email relevant data to users, and display department-specific information.

Field Name	Data Type	Description
userID (Primary Key)	Integers	Uniquely identifies the client
username	String	Helps department heads, admins etc see the name of each user.
password	String	Prevents unauthorised access to sensitive data, protects data integrity of users
role	String	Helps direct users to the appropriate webpage and display relevant data
email	String	Used to send budget reports, spending updates and status updates to users.
department_id	Integers	Determines which department each user belongs to.

Sample					
userID	username	password	role	email	department_id
4	DaniellaBurke	\$2b\$10\$MljOGkO6hboGcG47sM2oe	General Employee	danielburke@gmail.com	3
29	RoseWilliams	\$2b\$10\$qxsRtoYQXISl4epnGqgA	Department Head	williamsrose22@gmail.com	2
34	JackParker	\$2b\$12\$a/K07LpINZGi3	Admin	jackparker08@gmail.com	4

		M.JdrHy0u.lunyh3A.			
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Note : passwords are stored in their hashed values.

Departments Table :

The database stores information about each department, including its budget requests, associated Department Head, administrator comments, and approval status. This structure ensures that data is organised, accurate, and editable at the department level without affecting other records.

Field name	Data type	Description
Department_id (Primary Key)	Integers	Uniquely identifies each department
department_name	String	Helps department heads, admins and employees see the name and role of each department
requested_budget	Integers	Stores the amount requested by each department to shape admin feedback, budget editing and expenditure priorities.
budget_status	Characters	Stores the status (approval/denial/pending) of each budget so that department heads can track progress.
admin_comments	String	Stores admin feedback for each department's budget.
Department_head_id (Foreign Key)*	Integers	Stores the user_id of the department head in order to authenticate users, send update emails to the right user, etc.

* The department_head_id column is a foreign key referencing the user_id column in the Users table, but uses a different name as an alias.

Sample					
department_id	department_name	requested_budget	budget_status	admin_comments	department_head_id
2	HR	200,000	D	Miscellaneous expenditure not accounted for	29
3	Finance	500,000	A	Changes implemented; budget is well-planned	29 87
4	IT	40,000	P		3

Expenditures Table :

This table records individual spending entries submitted by Department Heads, including the expenditure name, amount, date, and associated department. By linking each record to a department through a foreign key, the system ensures that financial data remains organised, traceable, and department-specific, allowing administrators to monitor and compare actual spending against approved budgets.

Field Name	Data Type	Description
Expenditure_id (Primary Key)	Integers	Uniquely identifies each expenditure
expenditure_name	String	Provides context about the purpose of the expenditure
expenditure_amount	Integers	Stores the amount spent
date	Date	Stores the date of the expenditure for future reference
Department_id (Foreign Key)	Integers	Stores the department that made the expenditure

Sample				
Expenditure_id	expenditure_name	expenditure_amount	date	department_id
3	New computers for all employees	50,000	2025-02-01	Changes implemented; budget is well-planned
15	Employee work trip	20,000	2025-04-11	Miscellaneous expenditure not accounted for
24	Legal consultation	4,500	2025-08-07	

Test Plan

TEST NO.	SUCCESS CRITERIA	CRITERION FOR SUCCESS	TEST TO BE DONE	EXPECTED OUTCOME
1	A	Authentication features providing role-based access control (Admin, Department Head, Budget Controller).	1. Enter admin username (admin1) and password (admin123)	User should be redirected to admin functions page.
2	A	Authentication features providing role-based access control (Admin, Department Head, Budget Controller).	1. Enter department head username (dephead1) and password (dephead123)	User should be redirected to department head functions page.
3	A	Authentication features providing role-based access control (Admin, Department Head, Budget Controller).	1. Enter general employee username (user1) and password (user123)	User should be redirected to general employee functions page.
4	D	Test to see if a user can log in with correct credentials	1. Enter correct username (admin) and password (admin123). 2. Click "Login".	User should be logged in successfully and directed to the dashboard.

5	D	Test login with incorrect password.	1. Input correct username (admin) but incorrect password (admin472). 2. Click "Login".	User sees an error message: "Incorrect username or password" and is not logged in.
6	D	Test to see if the system prevents login when username and password fields are empty	1. Leave the username and password fields blank. 2. Click "Login".	User should receive an error message (e.g., "Username and password cannot be empty"), and login should be prevented.
7	C	All passwords should be hashed.	1. Register a new user (user1). 2. Access the database and check the stored password. In this case, the password was stored as \$2b\$12\$a/K07LpINZGi3M.JdrHy0u.lunyhx3A.gTGgqOsPU.5TGB4SiBKBi, which is a hashed value.	The password should be stored as a hashed value, not as plaintext.
8	C	All passwords should be hashed.	1. Reset the user password (for user head1, reset the password from head123 to head234). 2. Check the database for the new stored password.	The new password should be stored as a hashed value, not in plaintext. In this case, the password was stored as \$2b\$10\$/gXBsgZJJUOhKfhJQ.ELbuxXRRGAH8IO8fnqrfnX83o07v3O6LPj, which is a hashed value. Also, the hashed value of the new password should be different from the hashed value of the old password.
9	C	All passwords should be hashed.	1. Enter the correct username and password (username head1, password head234). 2. Verify that login is successful only if the password matches the hashed value.	System should only allow login if the entered password matches the hashed version stored in the database.

10	E	Admins can view budget plans vs. actual budget implementations for each department. Admins should be able to view planned vs. actual budget implementations for each department in the form of graphics.	1. Log into the admin dashboard page. 2. Verify that the bar graph depicts budget discrepancies accurately.	Budget table and planned vs actual graph displayed correctly. Bar graph visually and mathematically accurate. Budgets table has accurate data.
11	E	Admins can view budget plans vs. actual budget implementations for each department.	1. Verify that the total amount calculated for amount_spent is accurate.	Amount_spent by each department displayed on the graph should be the mathematically accurate sum of expenditure.
12	G	Admins should have a clear view of budget data and status for each department.	1. Verify that the table with departmental budget data is completely accurate and up-to-date.	Budgets table has accurate data.
13	G	Admins should have a clear view of budget data and status for each department.	1. Edit the budget proposal or comments for a certain department. 2. Verify that those same changes are made in the budget table.	Budget table should contain the new, updated values for that specific department.
14	I	Admins can edit departmental budget requests.	1. Log in with admin user_id (admin1) and password (admin123). 2. Click the edit button on a certain department's budget request (e.g., Accounts). 3. Change amount (e.g., \$6000 to \$10000). 4. Click the save changes button.	Amount of the department's budget request should be changed.

15	I	Admins can edit departmental budget requests.	1. View the budget and other details for the specific department from the budgets table of the database. 2. Verify that the amount is consistent with the new value entered by the admin.	Amount should be consistent with new value entered by admin.
16	M	Admins can add comments to proposed budget plans for each department.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select the department which the admin would like to add a comment to (e.g., Accounts). 3. Enter the comment. 4. Click the 'save changes' button.	The admin's comment should be stored in the comments column of the budget table in the database, for that specific department.
17	M	Admins can add comments to proposed budget plans for each department.	1. View the budget and other details for the specific department from the budgets table of the database. 2. Verify that the comment for the department is consistent with the new comment entered by the admin.	Comment should be consistent with new value entered by admin.

18	I	Admins can review, approve, or reject budget adjustments.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a department with pending budget status (e.g., Accounts) and click edit. 3. Enter an A to approve the budget proposal.	Budget_status column should be updated in the database to hold 'A' for approved.
19	I	Admins can review, approve, or reject budget adjustments.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a department with pending budget status (e.g., Accounts) and click edit. 3. Enter a D to deny the budget proposal.	Budget_status column should be updated in the database to hold 'D' for denied.
20	I	Admins can review, approve, or reject budget adjustments.	1. Log in with admin user_id (admin1) and password (admin123). 2. Verify that the table displayed includes departmental data of both approved, rejected and pending budget requests.	Table should be up-to-date.
21	F	Admins can search for and filter the department name and role of a certain employee.	1. Log in with admin user_id (admin1) and password (admin123). 2. Open the search bar. 3. Enter the name of an employee (e.g., user1).	Search results should display all data such as user_name, user_role and user_department for the selected employee.

22	F	Admins can search for and filter the department name and role of a certain employee.	1. Log in with admin user_id (admin1) and password (admin123). 2. Open the search bar. 3. Enter a certain role (e.g., department heads).	Search result should show user_name and department_name, etc. of all employees with the role of department heads.
23	F	Admins can search for and filter the department name and role of a certain employee.	1. Log in with admin user_id (admin1) and password (admin123). 2. Open the search bar. 3. Enter a certain department (e.g., Accounts).	Search result should show user_name and department_name user_role, etc. of all employees belonging to the selected department_name.
24	H	Admins can create, edit, and delete user accounts.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a certain user. 3. Click 'edit' and change the department the user belongs to (e.g., from HR to Accounts).	The database and user table displayed should show the new and updated department_name of the selected user.
25	H	Admins can create, edit, and delete user accounts.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a certain user. 3. Click 'delete' and save changes.	The database and user table displayed should not contain any data of the selected user.
26	H	Admins can create, edit, and delete user accounts.	1. Log in with admin user_id (admin1) and password (admin123). 2. Click the 'add user' button. 3. Add all user details.	The database and user table displayed should contain all details and data of the new user added.

27	J	Department Heads can edit budget request amounts for their department (in accordance with real-time occurrences).	1. Log in with department_head id (dephead1) and password (dephead123). 2. Click the 'edit' button. 3. Change the amount of the budget proposal (e.g., from \$8000 to \$6000).	The database and budget table displayed (with budget_status, etc.) should show the updated value of the proposed amount.
28	K	Department Heads can submit purchase requests for approval.	1. Log in with department_head id (dephead1) and password (dephead123). 2. Put the proposed amount of your budget in the column. 3. Verify that admin has received a notification (email) about the budget proposal.	Once the department heads add a budget request, it is sent to the admin for approval or denial.
29	L	Department Heads can track their submitted requests and view approval status.	1. Log in with department_head id (dephead1) and password (dephead123). 2. Verify that the budget status table displayed has accurate data about the amount requested, budget_status and department_name.	The budget table displayed should show accurate and up-to-date data.

30	L	Department Heads can track their submitted requests and view approval status.	1. Log in as admin, view the budget proposal of the department head (dephead1). 2. Approve the budget proposal (A).	The budget table for department heads should get updated to show approval status as 'A'.
31	E	Admins can track spending patterns and discrepancies.	1. Log in as admin, view the budget proposal of the department head (dephead1). 2. Ensure that visual representations and the budget table are displayed on the admin dashboard page with accurate data.	Budget table and planned vs actual graph displayed correctly. Bar graph visually and mathematically accurate. Budgets table has accurate data.
32	O	Admins can generate yearly budget reports in Excel or PDF format.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a specific department and click the "email to me" button so that the report gets emailed to the admin. 3. Check email for the downloaded file.	The downloaded pdf should be received over email (the correct email for the user) and should be in a readable format.

33	O	Admins can generate, export, and email comprehensive departmental budget reports to themselves.	1. Change admin email_id in the database (for the user admin1). 2. Select a specific department and click the "email to me" button so that the report gets emailed to the admin. 3. Check email for the downloaded file.	The file should be sent to the new email updated by the user.
34	P	General employees should be able to view all budget data relevant to their role.	1. Log in with general employee user_id (user1) and password (user123). 2. Ensure that the budget table displayed has accurate and up-to-date data.	Budget table displayed has mathematically and visually accurate data.
35	P	General employees should be able to view all budget data relevant to their role.	1. Change the department_head name in the budgets table of the same department as the general user. 2. Verify that the new name is updated in the budgets table displayed.	Data that was changed is updated in the budgets table and displayed accordingly.

36	Q	Department heads should be able to add expenditures to a spending table whenever they make a purchase.	1. Log in with department_head id (dephead1) and password (dephead123). 2. Click the 'add expenditure' button in the spending tracker and add an amount (e.g., \$200). 3. Add the 'expenditure purpose' (e.g., purchasing company laptops).	Data that was added to the spending tracker by the department head is updated to the spending table of the database.
37	Q	Department heads should be able to add expenditures to a spending table whenever they make a purchase.	1. Access the database and view the spending table for the department_head's specific department. 2. Verify that the expenditure_amount just entered is updated in the database.	Every expenditure made by the department head is updated in the database along with amount and purpose.

38	R	Real time budget tracking for Admins with notifications of important department expenditures over email.	1. Log in with department_head id (dephead1) and password (dephead123). 2. Click the 'add expenditure' button in the spending tracker and add an amount (e.g., \$200). 3. Add the 'expenditure purpose' (e.g., purchasing company laptops). 4. Verify that an email is sent to the admin's email id notifying him/her of the expenditure.	Email is sent to the correct email id with the correct data.
39	S	Onboarding process: Admins can create and assign roles to new users, and users receive an onboarding guide explaining system functionalities.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a new user and add the role as admin. 3. Verify that the user receives an onboarding guide customised to their role and explaining system functionalities.	User receives the admin onboarding guide, detailing their responsibilities and the features/functions they're allowed to access.

40	S	Onboarding process: Admins can create and assign roles to new users, and users receive an onboarding guide explaining system functionalities.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a new user and add the role as department head. 3. Verify that the user receives an onboarding guide customised to their role and explaining system functionalities.	User receives the department_head onboarding guide, detailing their responsibilities and the features/functions they're allowed to access.
41	S	Onboarding process: Admins can create and assign roles to new users, and users receive an onboarding guide explaining system functionalities.	1. Log in with admin user_id (admin1) and password (admin123). 2. Select a new user and add the role as general employee. 3. Verify that the user receives an onboarding guide customised to their role and explaining system functionalities.	User receives the general employee onboarding guide, detailing their responsibilities and the features/functions they're allowed to access.
42	V	Whenever a budget request is submitted and pending approval, admin should get a notification.	1. Submit a budget_amount and budget_purpose for a specific department. 2. Verify that admin gets a notification on his/her email id as registered in the database.	Admin should receive email about budget request with the correct data and on the correct email id.

43	U	Whenever a budget is approved or rejected, users and department heads should get an email notification.	1. Log in with admin user_id (admin1) and password (admin123). 2. Approve a budget request for a particular department. 3. Verify that department head and employees of that department received a notification on their email ids as registered in the database.	Department head and general employees should receive email about budget status with the correct data and on the correct email id.
44	B	Implement session management to maintain active user sessions securely, ensuring users remain authenticated while performing actions within the system. (Admin)	1. Log in using admin credentials (admin1, admin123). 2. Access multiple restricted pages (e.g., Admin Dashboard, Budget Approval, User Management). 3. Perform authorized actions such as editing or approving a budget. 4. Refresh the page or navigate between modules. 5. Attempt to access the same pages after manually deleting session cookies or logging out.	Admin should stay logged in across pages during an active session. Upon logout or session deletion, access to restricted pages should be denied and redirect to the login screen.

45	B	Implement session management to maintain active user sessions securely, ensuring users remain authenticated while performing actions within the system. (Department Head)	1. Log in using Department Head credentials (dephead1, dephead123). 2. Navigate between pages such as Department Budget and Expenditures. 3. Perform actions like submitting or viewing a budget request. 4. Close and reopen the tab within session timeout period. 5. Attempt to access admin-only pages via URL manipulation.	Department Head should remain authenticated within their session and have consistent access to authorized features only. Attempts to access admin-only pages should be denied.
46	B	Implement session management to maintain active user sessions securely, ensuring users remain authenticated while performing actions within the system. (General Employee).	1. Log in using General Employee credentials (user1, user123). 2. Submit a budget input form and view submission confirmation. 3. Navigate to other accessible sections. 4. Leave the session idle for a defined period (e.g., 10 minutes) and then interact again. 5. Log out and try revisiting a restricted URL using browser history.	Employee should remain logged in within the active session. After logout or timeout, session should expire and prevent access to any restricted URLs.